



Adhoc Sensor Networks UNIT2

Adhoc networks (Swami Vivekananda Institute of Technology)



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UNIT –II

DATA TRANSMISSION IN MANETS

UNIT-II

Data Transmission- Broadcast Storm Problem, **Rebroadcasting Schemes**-Simple-flooding, Probability-based Methods, Area-based Methods, Neighbor Knowledge-based: SBA, Multipoint Relaying, AHBP. **Multicasting:** **Tree-based:** AMRIS, MAODV; **Mesh-based:** ODMRP, CAMP; **Hybrid:** AMRoute, MCEDAR.

1. DATA TRANSMISSION

In general, any one of the mobile hosts has to broadcast the information to all its neighbors during data transmission. For far-away devices, the message is rebroadcasted which could cause collision if multiple devices broadcast at the same time and are in the neighborhood. This is also known as the *broadcasting storm problem*; we will discuss ways to perform efficient broadcasting of messages.

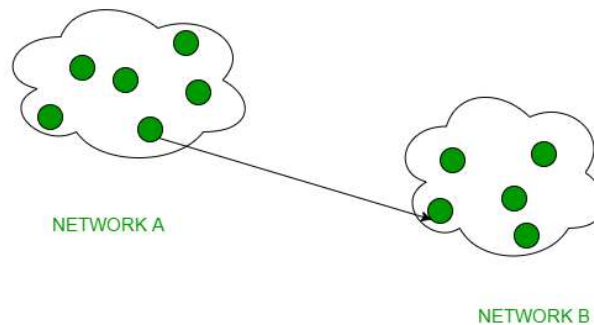
The MHs in the MANET share a single common channel with carrier sense multiple access (CSMA), but no collision detection (CD) capability (e.g., the IEEE standard 802.11 [IEEE-802.11.1997]). Synchronization in such a network with mobility is unlikely, and global network topology information is unavailable to facilitate the scheduling of a broadcast. Thus, one straightforward and obvious solution is to achieve broadcasting by flooding (for example, as it is done by mostly all MANET routing algorithms). Unfortunately, it is observed that redundancy, contention, and collision could exist if flooding is done blindly. Several problems arise in these situations including:

- As the radio propagation is omnidirectional (All directions) and a physical location may be covered by the transmission ranges of several hosts, many rebroadcasts are considered to be redundant.
- Heavy contention could exist because rebroadcasting hosts are close to each other; and
- As the RTS/CTS handshake is inapplicable for broadcast transmissions, collisions are more likely to occur as the timing of rebroadcasts is highly correlated.

a. UNICAST

This type of information transfer is useful when there is a participation of a single sender and a single recipient. So, in short, you can term it a one-to-one transmission. For example, if a device

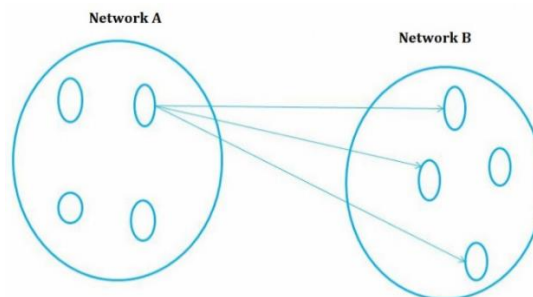
having IP address 10.1.2.0 in a network wants to send the traffic stream (data packets) to the device with IP address 20.12.4.2 in the other network, then unicast comes into the picture. This is the most usual form of data transfer over networks.



b. MULTICAST

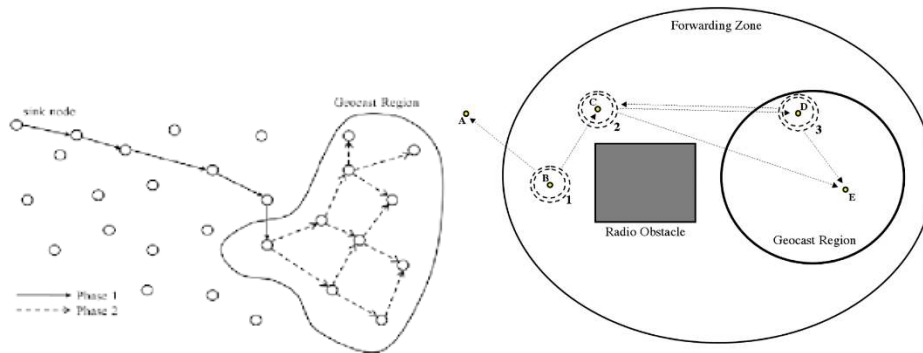
When the data is transmitted from a single source host to a specific group of hosts having the interest to receive the data, it is known as multicast transmission. Multicast can be more efficient than unicast when different groups of receivers need to see the same data.

Example – Multicast is the technique used in Internet streaming of video or audio teleconference, sending an email to a particular group of people, etc.



c. GEOCASTING

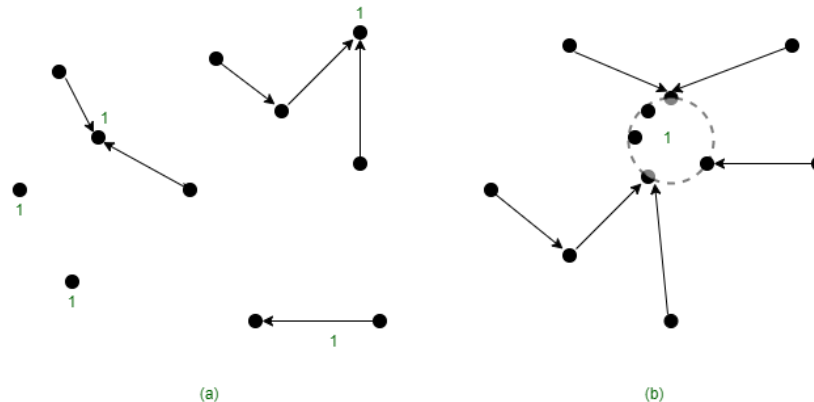
In a geocasting problem in ad hoc networks, a message is sent from one node to all the nodes located in a designated region. For example, monitoring center needs to contact all active sensors within a monitored area to either gather data from them periodically, or to provide its location to sensors covering certain area for event reporting. Intelligent flooding methods exist for this task when all active sensors belong to the monitored area. However, when a particular area containing only a small subset of active sensors needs to be monitored, the problem reduces to geocasting.



d. ANYCASTING

Anycast is a method for routing network traffic where the sender distributes packets to a destination that is adjacent to it in terms of network topology. The features of Anycasting is that the networking approach can allow for messages to be shared to a team of receivers that all have a similar destination address.

Suppose we want to anycast to the members of group 1. They will be given the address “1”, instead of different addresses. Distance vector routing will distribute vectors as usual, and nodes will choose the shortest path to destination 1. This will result in nodes sending to the nearest instance of destination 1. That is, it believes that all the instances of node 1 are the same node, as in the topology.



e. BROADCASTING

In Broadcast transmission, the data is transmitted from one or more senders to all the receivers within the same network or in other networks. This type of transmission is useful in network management packets such as ARP (Address Resolution Protocol) and RIP (Routing Information Protocol) where all the devices must see the data.

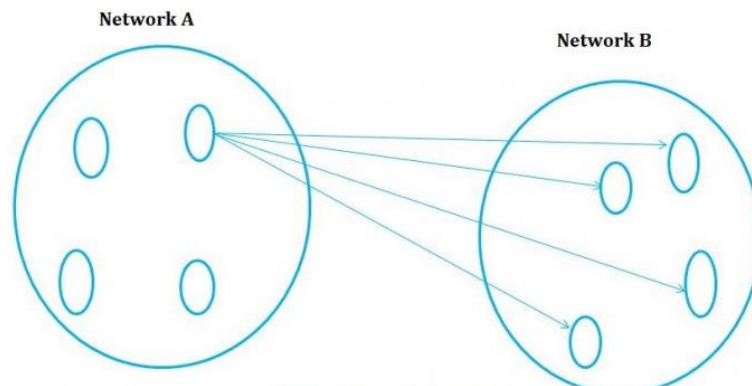
There are two types of broadcast transmission –

- Directed Broadcast, and
- Limited Broadcast

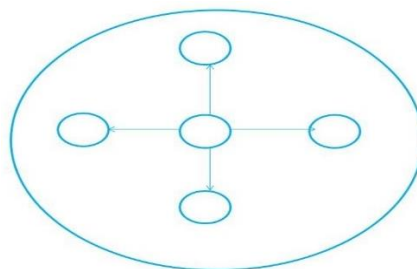
Directed Broadcast transmits data from one source host to all the other hosts that exist in some other

network. It is used in two scenarios –

- When the hosts are responsible for parsing data from broadcast packets.
- When all the hosts require the same data.



In Limited Broadcast, the data is transmitted from a single source host to all the other hosts residing in the same network. Suppose we have to send a stream of packets to all the devices over the network that you reside, this broadcasting comes in handy. For this to achieve, it will append 255.255.255.255 (all the 32 bits of IP address set to 1) called Limited Broadcast Address in the destination address of the datagram (packet) header which is reserved for information transfer to all the recipients from a single client (sender) over the network.



f. BROADCAST STORM PROBLEM

A broadcast storm is an abnormally high number of broadcast packets within a short period of time.

They can overwhelm switches and endpoints as they struggle to keep up with processing the flood of packets. When this happens, network performance degrades, i.e. A broadcast storm occurs when a network system is overwhelmed by continuous multicast or broadcast traffic. When different nodes are sending/broadcasting data over a network link, and the other network devices are rebroadcasting the data back to the network link in response, this eventually causes the whole network to melt down and lead to the failure of network communication. There are many reasons a broadcast storm occurs, including poor technology, low port rate switches and improper network configurations. A broadcast storm is also known as a network storm.

For example, suppose there is a small LAN network consisting of three switches (Switch A, Switch

B and Switch C), and three network segments (Segment A, Segment B and Segment C). Two nodes are attached within this network. Node A is attached to Segment B, while Node B is directly attached to Switch A. Now, if Node B wants to transmit a data packet to Node A, then traffic is broadcast from Switch A over to Segment C; if this fails, then Switch A also broadcasts traffic over Segment A. Because Node A neither attaches to Segment C, nor Segment A, these switches would further create a flood to Segment B. If neither device/switch has learned the Node A address, then traffic is sent back to Switch A. Hence, all devices/switches keep sending and resending the traffic, eventually resulting in a flood loop or broadcast loop. The final result is that the network melts down, causing failure in all network links, which is referred to as a broadcast storm.

The following elements play an active role in the creation of a broadcast storm:

- Poor network management
- Poor monitoring of the network
- The use of cheap devices, including hubs, switches, routers, cables, connectors, etc.
- Improperly maintained network configuration and inexperienced network engineers
- The lack of a network diagram design, which is needed for proper management and to provide guidelines for all network traffic routes. This can be done on paper and with the help of application software that creates an automated network diagram.

How to reduce broadcast storms

- **Storm control and equivalent protocols** allow you to rate-limit broadcast packets. If your switch has such a mechanism, turn it on.
- **Ensure IP-directed broadcasts are disabled on your Layer 3 devices.** There is little to no reason why you would want broadcast packets coming in from the internet going to a private address space. If a storm originates from the WAN, disabling IP-directed broadcasts will shut it down.
- **Split up your broadcast domain.** Creating a new VLAN and migrating hosts into it will load and balance the broadcast traffic to a more acceptable level. Broadcast traffic is necessary and useful, but too much of it eventually leads to poor network experience.
- **Check how often ARP tables are emptied.** The more frequently they are emptied, the more often ARP broadcast requests occur.
- **Sometimes, when switches have a hardware failure,** their switch ports begin to spew out broadcast traffic onto the network. If you have a spare switch of the same or similar model, clone the config of the active switch onto the spare and swap the hardware and cables during

a maintenance window. Does the storm subside? If it does, it was a hardware issue. If not, then you've to keep digging.

- **Check for loops in switches.** Say there was an unmanaged Layer 2 switch connected upstream to an unmanaged switch, and someone is connected a cable between two ports on the same unmanaged switch (let us say ports 1 and 2). The unmanaged switch will respond to all broadcasts multiple times and flood the broadcast domain with packets, causing a denial-of-service attack on the network.
 - **BPDU and PortFast** or equivalent features should be implemented as a best practice to prevent loops.
 - Discourage users from connecting unmanaged switches to managed switch ports by enforcing a maximum number of MAC addresses per port. This may be up to two MAC addresses if users have a computer plugged into an IP phone, which in turn is plugged into the switch.

2. REBROADCASTING SCHEMES

Broadcasting is the simple and basic process in Mobile Ad Hoc Networks in which the same packet has been transmitted from the sender node to all the remaining nodes of the network. Due to the limited radio range of mobile nodes, the MANET is multihop in nature. Hence, the packet which is transmitted from the valid mobile source cannot reach the target in a single hop. So, here some other nodes of the network are required to forward the source transmitted packet to the destination. These nodes are often known as intermediate nodes. It can rebroadcast the packet to enable its delivery when the destination is in communication range through the intermediate nodes. This process is commonly known as "packet forwarding" or "flooding." In general, the broadcasting strategies can be grouped into four families: Simple flooding, Probability-based methods, Area-based methods, and Neighbor knowledge-based methods.

3. SIMPLE-FLOODING

In this method, a sender node initiates a message to all its neighbors. Each of these neighbors will check if they have seen this message before, if yes, the message will be dropped, if not the message will rebroadcasted at once to all their neighbors. The process goes on until all nodes have the message. This method is suitable for MANET with low density nodes and high mobility. It ensures no packet losses. But it may cause network congestion and quickly drain the battery power. Blind flooding ensures coverage; the broadcast packet is guaranteed to be received by every

node in the network. Redundant transmissions in blind flooding may cause the broadcast storm problem, in which redundant packets cause contention and collision.

a. PROBABILITY-BASED METHODS

Probability Based Flooding

When a node receives a broadcast message for the first time, the node rebroadcasts the message with a probability P . If the message received is already seen, then the node drops the message irrespective of whether the node retransmitted the message when received for the first time. Thus, randomly having some nodes not rebroadcast saves node and network resources without harming delivery effectiveness. Probabilistic broadcasting is one of the simplest and most efficient broadcast techniques. In this approach, each intermediate node rebroadcast received packets only with a predetermined forwarding probability. When the probability reaches 100% then it is identical to simple flooding.

Counter Based Methods

An inverse relationship between the number of times a packet is received at a node and the probability of that node being able to reach additional area on a rebroadcast. This result is the origin of their Counter-Based scheme. Upon reception of a broadcast packet, the node initiates a counter with a value of one and sets RDT (Random Delay Timer). During the RDT, the counter is incremented by one for each redundant packet received. If the counter is less than a threshold value when the RDT expires, the packet is rebroadcast. Otherwise, it is simply dropped.

b. AREA-BASED METHODS

Distance based approach

In Distance based approach, a node compares the distance between itself and each neighboring node that has previously forwarded a given packet. Upon reception of a previously unseen packet, a Random Assessment Delay (RAD) is initiated, and redundant packets are cached. When the RAD expires, all source node locations are examined to see if any node is closer than a threshold distance value. If true, the node does not rebroadcast. So, a node using the distance-based approach needs the information of the geographic locations of its neighbors to make a rebroadcast decision. Measuring the distance of the source of the received packet may be accomplished by physical layer parameter i.e., signal strength at the node. Otherwise, if a GPS receiver is available, the location information can be included in each packet of the nodes that are transmitted.

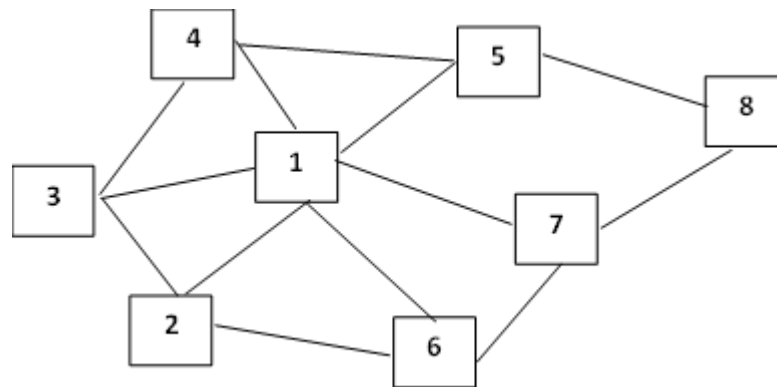
Location based Method

In this method each node has to identify its own location relative to the location of the sender using the geo location technique e.g., Global Positioning System. Each node in a MANET will add its own location to the header of each message it sends or rebroadcasts. When a neighboring node receives the packet, it notes the location of the sender and computes the additional coverage area obtainable if it were to rebroadcast. If the additional coverage area to rebroadcast is less than the given threshold, the node will not rebroadcast, and the same packets are ignored. Otherwise, the node assigns an RDT before delivery. During RAD, a redundant packet is received by a node then it is recomputed to the additional coverage area and compares that value to the threshold. The comparison of the area calculation and threshold occurs for all redundant broadcasts received until the packet reaches either the scheduled send time or is dropped.

c. NEIGHBOR KNOWLEDGE-BASED

Self-Pruning

Self-Pruning is the simplest method in reducing broadcast redundancy. In this method, each node requires knowledge of its 1-hop neighbors, which is acquired by sending periodic “Hello” packets. A node contains a list of known neighbors in the header of each broadcast packet. Broadcast packet received by neighbor nodes checks whether their list matches Additional nodes, it rebroadcasts else it avoids doing rebroadcasting. In the below figure, after receiving a message from node 6 node 2 will rebroadcast the message to node 3 and node 1 as its only additional nodes. Note that node 1 also will rebroadcast the same message to node 3 as its only additional node. In this situation still the message redundancy takes place.



Dominant Pruning

Dominant pruning also uses two-hop neighbor knowledge, obtained via hello packets, for routing decisions. Unlike SBA, however, dominant pruning requires rebroadcasting nodes to proactively choose some or all its one-hop neighbors as rebroadcasting nodes. Whenever a node receives a broadcast packet, it checks the header to see if its address is a part of the list. If so, it now must determine which of its neighbors should rebroadcast its packet to include them in the packet header. For that, it uses a Greedy Set Cover algorithm given the knowledge of which neighbors have already been covered by the sender's broadcast. One such algorithm recursively chooses one-hop neighbors that cover all two-hop neighbors.

Multi-point Relaying

Multipoint Relaying is like Dominant Pruning; upstream senders will choose the rebroadcasting nodes. For example, Node A is instigating a broadcast packet. It will select some or all one hop neighbor nodes to rebroadcast packet that they receive from Node A. The chosen nodes are called Multipoint relays (MPRs). Each MPR is necessary to decide a subset of its one hop neighbors to act as MPRs. Since a node knows the network topology within a 2-hop radius, it can select 1-hop neighbors as MPRs that most efficiently reach all nodes within the two-hop neighborhood. For a node to choose its MPRs:

- 1-hop neighbor must discover all 2-hop neighbors that can be reached by them. 1-hop neighbors must be allotted as MPRs.
- Decide the resultant cover set (i.e., the collection of 2-hop neighbors that will receive the packet from the current MPR set).
- From the left over 1-hop neighbors not yet in the MPR set, determine the one that would cover the majority 2-hop neighbors not in the cover set.
- Till all 2-hop neighbors are covered repeat step 2.

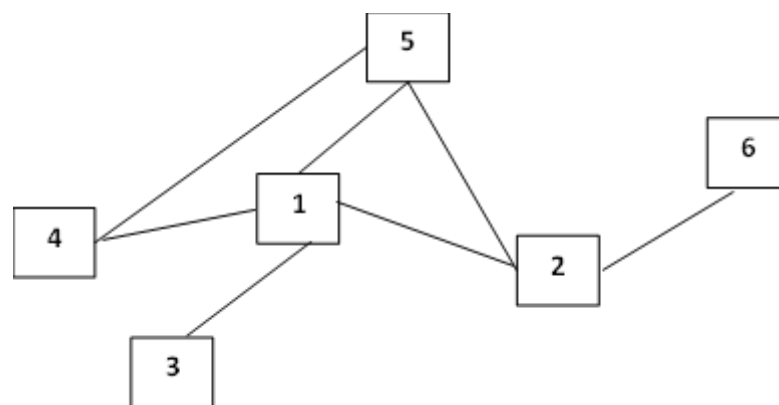
SBA

The Scalable Broadcast Algorithm (SBA) requires all the nodes to have knowledge about neighbors with two hop radius. With the neighbor knowledge, nodes determine whether it could reach additional nodes by rebroadcasting. By sending periodically “hello” packets, 2-hop neighbor knowledge is attainable; each “hello” packets have identifier of node and the list of known neighbors. Two hop radius node information are known by nodes after receiving “Hello” packets from neighbor nodes. Suppose Node A sends a broadcast packet to Node B. Node B knows all neighbors of Node A since it is a neighbor. Other than Node A’s broadcast if Node B has additional neighbors, it prepares the packet for delivery with a RAD. suppose redundant broadcast packet from another neighbor are received by Node B, it finds out whether can reach by rebroadcasting.

AD HOC BROADCASTING APPROACH (AHBP)

In this approach, only nodes selected as gateway nodes and a broadcast message header are allowed to rebroadcast the message. The approach is described as follows:

- Locate all two hop neighbors that can only be reached by one hop neighbor. Select these one hop neighbors as gateways.
- Calculate the cover set that will receive the message from the current gateway set for the neighbors not yet in the gateway set, find the one that would cover the most two hop neighbors not in the cover set. Set this one hop neighbor as a gateway.
- Repeat process 2 and 3 until all two hop neighbors are covered. When a node receives a message and is a gateway, this node determines which of its neighbors already received the message in the same transmission.



In Ad hoc broadcasting approach (figure), node 2 has 1, 5 and 6 nodes as one hop neighbors, 3 and 4 nodes as two hop neighbors. Node 3 can be reached through node 1 as a one hop neighbor of node 2. Node 4 can be reached through node 1 or node 5 as one hop neighbors of node 2. Node 3 selects node 1 as a gateway to rebroadcast the message to nodes 3 and 4. Upon receiving the

message node 5 will not rebroadcast the message as it is not a gateway.

4. MULTICASTING

Multicasting is the transmission of packets to a group of hosts identified by a single destination address. Multicasting is intended for group-oriented computing (broadcast a message to a subset of MANET MHs). The members of a host group are dynamic, that is hosts may join and leave groups at any time. There is no restriction on the location or number of members in a host group. A host may be a member of more than one group at a time. A host does not have to be a member of a group to send packets to it.

Protocols, designed for fixed networks, may fail to keep up with node movements and frequent topological changes as MHs become increasingly mobile, these protocols need to evolve to cope with the new environment. But the host mobility increases the protocol overheads. The broadcast protocols cannot be used either as multicasting requires a selected set of nodes to receive the message while all multicast algorithms depend on the topology of the network and do not consider whether a node belongs to a group or not. With this approach, data packets are sent throughout the ad hoc network and every node that receives this packet broadcasts it to all its immediate neighbors' nodes exactly once.

We can classify the protocols into four categories based on how route to the members of the group is created:

- Tree-Based Approaches
- Meshed-Based Approaches
- Stateless Multicast
- Hybrid Approaches

a. TREE-BASED Approaches

Most of the schemes for providing multicast in wired network are either source-based or shared tree-based. Due to simplicity of tree structures, many characteristics can be identified such as: a packet traverses each hop and node in a tree at most once, very simple routing decisions at each node, and the number of copies of a packet is minimized,

tree structure built representing shortest paths amongst nodes, and a loop-free data distribution structure.

On the other hand, there are many issues that must be addressed in tree-based approaches. Trees provide a unique path between any two nodes. Therefore, having even one link failure could mean reconfiguration of the entire tree structure and could be a major drawback. In addition, multiple packets generated by different sources will require some consideration when utilizing multicast trees such that efficient routing can be established and maintained. Thus, it is common to consider the use of either a shared tree or establish a separate tree per each source

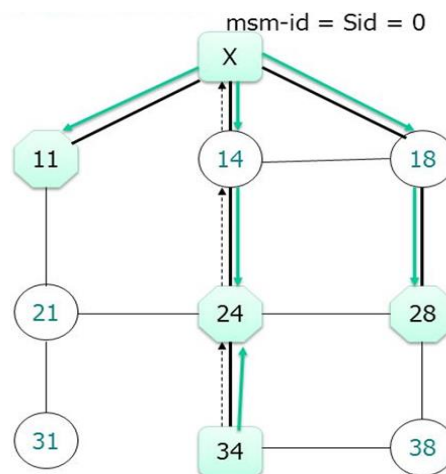
For separate source trees, each router or node involved in multiple router groups must maintain a list of pertinent information for each group in which it is involved. Such management per router is inefficient and not scalable. On the other hand, for shared trees, there is a potential that packets may not only traverse shorter paths, but in fact may be routed on paths with much longer distances than the shortest paths. While any scheme has positive and negative sides, the simple structure coupled with ease of approach has made multicast trees the primary method for realizing multicasting on the Internet.

i. AMRIS

Ad hoc Multicast Routing Protocol utilizing *Increasing id-numbers (AMRIS)* is an on-demand protocol, which constructs a shared multicast delivery tree to support multiple senders and receivers in a multicast session. AMRIS dynamically assigns an id- number to each node in each multicast session. Based on the id-number, a multicast delivery tree rooted at a special node with Sid (Smallest-ID) is created and the id-number increases as the tree expands from the Sid. Generally, Sid is the source or the node that initiates a multicast session.

The first step in AMRIS protocol operation is the selection of Sid. If there is only one sender for a group, the Sid is the source of the group. In the case of multiple senders, a Sid is selected from the given set of senders. Once a Sid is identified, it sends a *NEW-SESSION* message to its neighbors. The content of this message includes Sid's *msm-id* (multicast session member id) and the routing metrics. Nodes receiving the *NEW-SESSION* message generate their own *msm-ids*, which are larger than the *msm-id* of the sender. In case a node receives multiple *NEW-SESSION* messages from different nodes, it keeps the message with the best routing metrics and calculates its *msm-ids*. To join an ongoing session, a node checks the *NEW-SESSION* message, determines a parent with smallest *msm-ids*, and unicast

a *JOIN-REQ* to its potential parent node. If the parent node is already in the multicast delivery tree, it replies with a *JOIN-ACK*. Otherwise, the parent itself tries to join the multicast tree by sending a *JOIN-REQ* to its parent. If a node is unable to find any potential parent node, it executes a branch reconstruction (BR) process to rejoin the tree. BR consists of two sub-routines, namely, subroutines 1 (BR1) and 2 (BR2). BR1 is executed when a node has potential parent node for a group. In case it does not find any potential parent node, BR2 is executed. In BR2, instead of sending a unicast *JOIN_REQ* to a potential parent node, the node broadcasts a *JOIN-REQ* which consists of a range field R to specify the nodes till R hops. Upon link breakage, the node with larger *msm-id* tries to rejoin the tree by executing any of the BR mechanisms. It is to be noted that AMRIS detects the link disconnection by a beaconing mechanism. Hence, until the tree is reconstructed there is a possibility of packets being dropped.



X, 34 -> Receivers / Senders, 14, 18 --> I- Nodes, 11,24,28 --> Receivers, 21,31,38 --> U-Nodes

ii.MAODV

The *Multicast Ad hoc On-Demand Distance Vector (MAODV)* protocol enables dynamic, self-starting, multihop routing between participating mobile nodes wishing to join or participate in a multicast group within an ad hoc network. The membership of these multicast groups is free to change during the lifetime of the network. MAODV enables mobile nodes to establish a tree connecting multicast group members. Mobile nodes can respond quickly to link breaks in multicast trees by repairing these breaks in a timely manner. In the event of a network

partition, multicast trees are established independently in each partition, and trees for the same multicast group are quickly connected if the network components merge.

One distinguishing feature of MAODV is its use of sequence numbers for multicast groups. Each multicast group has its own sequence number, which is initialized by the multicast group leader and incremented periodically. Using these sequence numbers ensures that routes found to multicast groups are always the most current ones available. Given the choice between two routes to a multicast tree, a requesting node always selects the one with the greatest sequence number.

The AODV multicast algorithm uses similar RREQ and RREP messages as in unicast operation. The nodes join the multicast group on-demand, and a multicast tree is created in the process. The tree consists of the group members and nodes connected to the group members. This enables a recipient host to join a multicast group even if it is more than one hop away from a multicast group member. The unicast operation of the protocol also benefits from the information that is gathered while discovering routes for multicast traffic. This cuts down the signaling traffic in the network.

A MH originates an RREQ message when it wishes to join a multicast group, or when it has data to send to a multicast group, but it does not have a route to that group. Only a member of the desired multicast group may respond to a join RREQ

If the RREQ is not a join request, any node with a fresh route (based on group sequence number) to the multicast group may respond. If an intermediate node receives a join RREQ for a multicast group of which it is not a member or if it receives a RREQ and it does not have a route to that group, it rebroadcasts the RREQ to its neighbors.

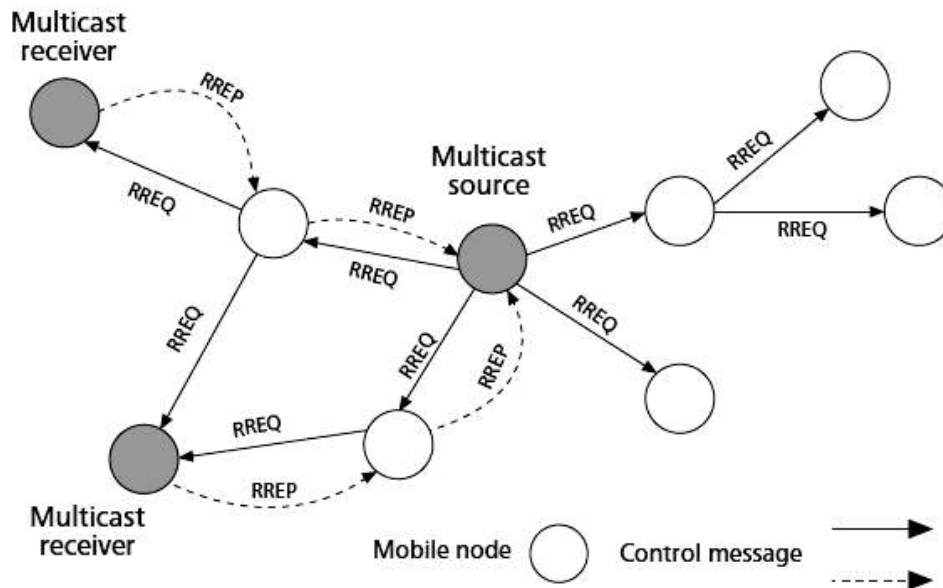
Route Request Message Format:

0 - 7	8	9	10	11 – 23	24 - 31
Type	J	R	G	Reserved	Hop Count
Other fields as in AODV					

J --> Join flag; set when source node wants to join a multicast group .

R--> Repair flag, set when a node is responding to a repair request to connect two previously disconnected portions of the multicast tree.

G--> Group Leader flag;



Route Discovery:

When a node wishes to find a route for a multicast group, it sends an RREQ message. The destination address in the RREQ message is set to the address of the multicast group. If the node wants to join the group in question, the J-flag in the message is set. Any node may respond to a RREQ merely looking for a route, but only a router in the desired multicast tree may respond to a join RREQ. The corresponding RREP message may travel through nodes that are not members of the multicast group. This means that the eventual route may also include hops through non-member nodes

Route Reply:

The multicast RREP message is slightly different from the unicast RREP. The address of the multicast group leader is stored in a field called Group-Leader-Addr. In addition, there is a field called Mgroup-hop. This field is initialized to zero and it is incremented at each hop along the route. Mgroup-hop contains the distance in hops of the source node to the nearest member of the multicast tree.

Because the protocol relies on a group-wide destination sequence number (DSN) to ensure fresh routes, the group leader broadcasts periodical Group Hello messages. The Group Hello is an unsolicited RREP message that has a TTL (Time to Leave) greater than the diameter of the network. The message contains extensions that indicate the multicast group addresses and corresponding sequence numbers of all the groups for which the node is the group leader. The sequence number for each group is incremented each time the *Group Hello* is broadcast. The *Hop_Cnt* field in the message is initialized as zero and incremented by the intermediate nodes.

The nodes receiving the Group Hello use the information contained therein to update their request tables. If a node does not have an entry for the advertised multicast group, one is inserted. The hop counts are used to determine the current distance from the group leader.

0 - 7	8	9 - 18	19 - 23	24 - 31
Type	R	Reserved	Prefix Size	Hop Count
Other fields as in AODV				

R--> Repair flag, set when a node is responding to a repair request to connect two previously disconnected portions of the multicast tree.

b. MESH-BASED Approaches

Mesh-based multicast protocols may have multiple paths between any source and receiver pairs. The tree-based protocols are not necessarily the best suited for multicast in a MANET environment if the network topology changes frequently. In such an environment, mesh-based protocols outperform tree-based proposals due to availability of alternative paths, which allow multicast datagrams to be delivered to the receivers even if links fail.

The disadvantage of a mesh is the increase in data-forwarding overhead. The redundant forwarding consumes more bandwidth in the bandwidth constrained ad hoc networks. Moreover, the probability of collisions is higher when a larger number of packets are generated. Therefore, one common problem in mesh-based protocols must consider is how to minimize the data-forwarding overhead caused by flooding. As we shall see, different protocols attack this issue in different ways using forwarding groups, cores, and so on. There are multiple protocols of the mesh-based approaches that support multicast in MANETs.

i. ODMRP

ODMRP (On-Demand Multicast Routing Protocol) is a mesh-based, multicast protocol that provides richer connectivity among multicast members. By building a mesh and supplying multiple routes, multicast packets can be delivered to destinations in the face of node movements and topology changes. In addition, the drawbacks of multicast trees in mobile wireless networks (e.g., intermittent connectivity, frequent tree reconfiguration, traffic concentration, etc.) are avoided. To establish a mesh for each multicast group, ODMRP uses the concept of forwarding group. The forwarding group is a set of nodes responsible for forwarding multicast data on shortest paths between any member pairs. ODMRP also applies OnDemand routing techniques to avoid channel overhead and improve scalability. No explicit control message is required to leave a group.

When a multicast source has packets to send, but no route to the multicast group, it broadcasts a *Join-Query* control packet to the entire network. This *Join-Query* packet is periodically broadcasted to refresh the membership information and updates routes. When an intermediate node receives a *Join-Query* packet, it stores the source ID and the sequence number in its message cache to detect any potential duplicates. The routing table is updated with an appropriate node ID (i.e., backward learning) from which the message was received. If the message is not a duplicate and the TTL is greater than zero, it is rebroadcasted.

When a *Join-Query* packet reaches a multicast receiver, it creates and broadcasts a *Join-Reply* to its neighbors. When a node receives a *Join-Reply*, it checks if the next hop node ID of one of the entries matches its own ID. If it does, the node realizes that it is on the path to the source and thus is a part of the forwarding group and sets the FG_FLAG (Forwarding Group Flag). It then broadcasts its own *Join-Reply* built upon matched entries. The next hop node ID field contains the information extracted from its routing table. In this way, each forward group member propagates the *Join-Reply* until it reaches the multicast source via the selected (shortest) path. This whole process constructs (or updates) the routes from sources to receivers and builds a mesh of nodes. After establishing a forwarding group and route construction process, a source can multicast packets to receivers via selected routes and forwarding groups. While a node has data to send, the source periodically sends *Join-Query* packets to refresh the forwarding group and the routes. When receiving the multicast data packet, a node forwards it only when it is not a duplicate and the setting of the FG_FLAG for the multicast group has not expired. This procedure minimizes the traffic overhead and prevents sending packets through stale routes.

In ODMRP, no explicit control packets need to be sent to join or leave the group. If a multicast source wants to leave the group, it simply stops sending *Join-Query* packets since it does not have any multicast data to send to the group. If a receiver no longer wants to receive from a particular multicast group, it does not send the *Join-Reply* for that group. Nodes in the forwarding group are demoted to non-forwarding nodes if not refreshed (no *Join-Replies* received) before they timeout.

ii. CAMP

The *Core-Assisted Mesh Protocol (CAMP)* supports multicasting by creating a shared mesh for each multicast group. Meshes thus created helps in maintaining the connectivity to the multicast users, even in case of node mobility. It borrows concepts from CBT (Core Based Tree Protocol), but the core nodes are used for control traffic needed to join multicast groups. The basic operation of the CAMP includes building and maintaining the multicast mesh for a multicast group. It assumes a mapping service, which provides routers with the addresses of groups identified by their names. Each router maintains a routing table (RT) built with the unicast routing protocol and is modified by CAMP when a multicast group needs to be inserted or removed. A router may update its MRT based on topological changes or messages received from its neighbors.

CAMP classifies the nodes in the network in three modes: simplex, duplex and non-member. A router joins a group in a simplex mode if it intends only to send traffic received from specific nodes or neighbors to the rest of the group and does not intend to forward packets from the group. A duplex member forwards any multicast packets for the group, whereas a non-member node needs not to be in the multicast delivery mesh. CAMP uses a receiver-initiated method for routers to join a multicast group. If a router wishing to join a group has multiple neighbors that are duplex members of the multicast group, then it simply changes its MRT and directly announces to its neighbors that it is a new member for the multicast group using multicast routing update. If it has no neighbors that are members of the multicast group, it either propagates a join request to one of the multicast group "cores" or attempts to reach a member through expanding ring search. Any router that is a regular member of the multicast group and has received the join request, is free to transmit a join acknowledgement (ACK) to the sending router. A router can leave a group if it has no hosts that are members of the group, and it has no neighbors for whom it is an anchor, i.e., if they are not needed to provide efficient paths for the dissemination of packets in the multicast meshes for the groups. Cores are also allowed to leave multicast group if there are no routers using them as anchors.

CAMP ensures that the mesh contains all reverse shortest paths between a source and the recipients. A receiver node periodically reviews its packet cache in order to determine whether it is receiving data packets from neighbors, which are on the reverse shortest path to the source. Otherwise, a HEARTBEAT message is sent to the successor in the reverse shortest

path to the source. This HEARTBEAT message triggers a PUSH JOIN (PJ) message. If the successor is not a mesh member, the PJ forces the specific successor and all the routers in the path to join the mesh.

CAMP has the advantage that it does not use flooding and the requests only propagate to mesh members. On the other hand, CAMP relies on an underlying unicast routing protocol to guarantee correct distances to all destinations within a finite time.

iii. HYBRID

The protocols to provide multicast in ad hoc networks discussed so far, either address efficiency or robustness but not both simultaneously. The tree-based approaches provide high data forwarding efficiency at the expense of low robustness, whereas mesh-based approaches lead to better robustness (link failure may not trigger a reconfiguration) at the expense of higher forwarding overhead and increased network load.

Thus, there is a possibility that a hybrid multicasting solution may achieve better performance by combining the advantages of both tree and meshed-based approaches. The different hybrid approaches which enable ad hoc multicasting are discussed below.

a) AMROUTE

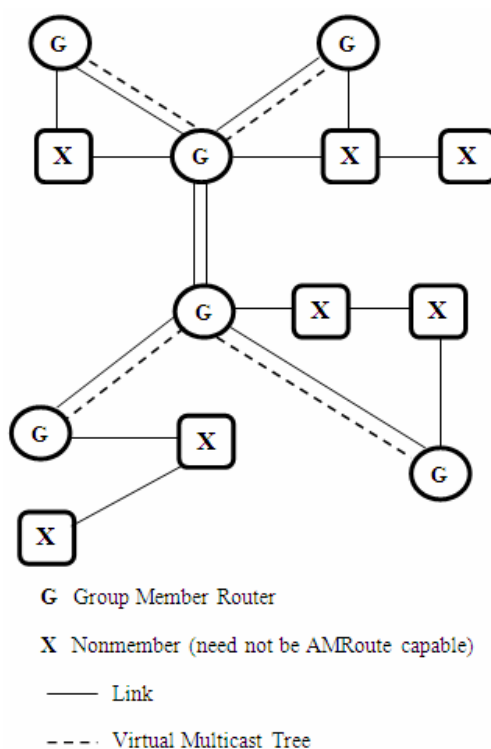
The Ad hoc Multicast Routing Protocol (AMRoute) creates a bi-directional, shared tree by using only group senders and receivers as tree nodes for data distribution. The protocol has two main components: mesh creation and tree setup.

The mesh creation identifies and designates certain nodes as logical cores, and these are responsible for initiating the signaling operation and maintaining the multicast tree to the rest of the group members. A non-core node only responds to messages from the core nodes and serves as a passive agent. The selection of logical core in AMRoute is dynamic and can migrate to any other member node, depending on the network dynamics and group membership. AMRoute does not address network dynamics and assumes the underlying unicast protocol to take care of it. To create a mesh, each member begins by identifying itself as a core and broadcasts JOIN_REQ packets with increasing TTL to discover other members. When a core receives JOIN_REQ from a core in a different mesh for the same group, it replies with a JOIN_ACK. A new bi-directional tunnel is created between the two cores and one of them is

selected as core after the mesh merger. Once the mesh has been established, the core initiates the tree creation process. The core sends out periodic TREE_CREATE messages along all links incident on its mesh. Using unicast tunnels, the TREE_CREATE messages are sent only to the group members. Group members receiving non-duplicate TREE_CREATE message forwards it to all mesh links except the incoming one and marks the incoming and outgoing links as tree links.

If a link is not going to be used as part of the tree, the TREE_CREATE is discarded and TREE_CREATE__NAK is sent back to incoming links. A member node, which wants to leave a group, can do so by sending a JOESLNAK message to its neighboring nodes.

AMRoute employs virtual mesh links to establish the multicast tree, which helps in keeping the multicast delivery tree the same even with the change of network topology as long as routes between cores (nodes) and tree members exist via mesh links. The main disadvantage of this protocol is that it may have temporary loops and may create nonoptimal trees in case of mobility.



b) MCEDAR

The *Multicast Core-Extraction Distributed Ad hoc Routing* (MCEDAR) is a multicast extension to the CEDAR architecture. The main idea of MCEDAR is to provide the efficiency of the tree-based forwarding protocols and robustness of mesh-based protocols by combining these two approaches. It is worth pointing out that a source-based forwarding tree is created on a mesh. As such, this ensures that the infrastructure is robust and data forwarding occurs at minimum height trees. MCEDAR decouples the control infrastructure from the actual data forwarding to reduce the control overhead.

The underlying unicast protocol, CEDAR, provides the core broadcasting for multicasting. The core is used for routing management and link state inspection. Also, the cores make up the mesh infrastructure, which is referred to as an mgraph, and use joinIDs to perform the join operation. As MCEDAR uses a mesh as the underlying infrastructure, it can tolerate a few link breakages without reconfiguration. The efficiency is achieved by using a forwarding mechanism on the mesh that creates an implicit route-based forwarding tree. As mentioned earlier, this ensures that the packets need to travel only the minimum distance in the tree.

